

relative distances



Gromov-Wasserstein: Formulation

- Let (X, d_X, α) and (Y, d_Y, β) be two metric measure spaces with distances $d_X(x_i, x_{i'}), d_Y(y_j, y_{j'})$

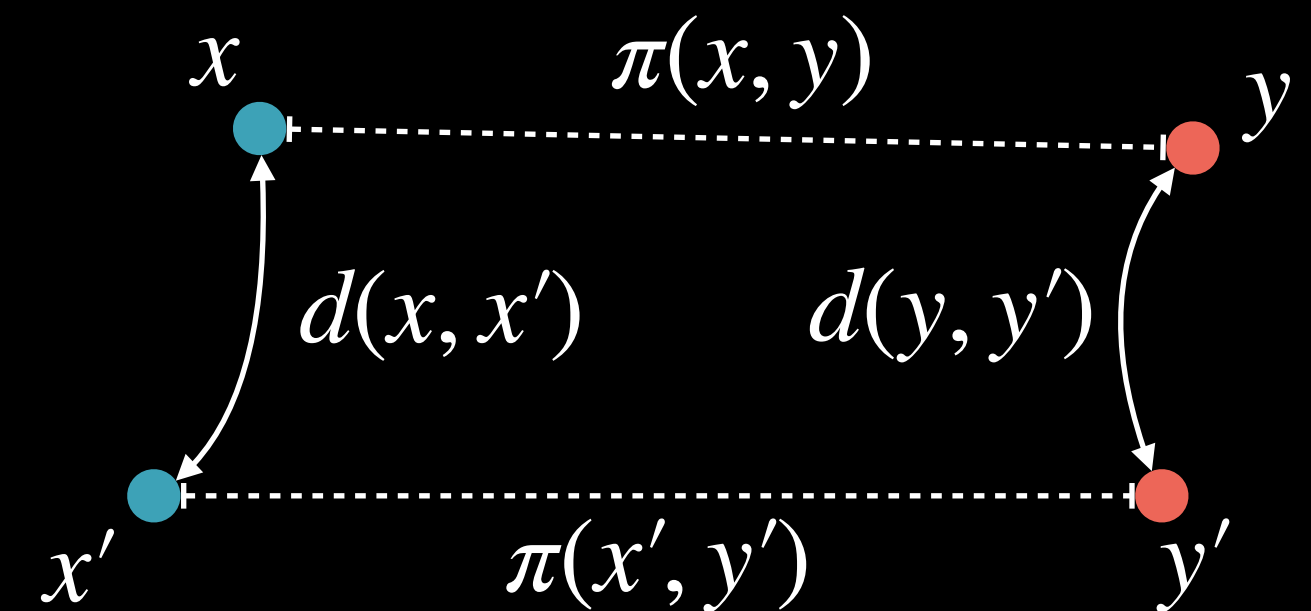
- The GW problem is:

$$\text{GW}((\alpha, d_X), (\beta, d_Y)) = \min_{\pi \in \Pi(\alpha, \beta)} \iint_{X^2 \times Y^2} \ell(d_X(x, x'), d_Y(y, y')) \, d\pi(x, y) \, d\pi(x', y')$$

where ℓ is a loss function, typically squared distance.

- Intuition:**

- Compares **relative distances** of pairs (x, x') and (y, y')
- Distances are weighted by coupling mass — π plays same role as before
- Minimizing it means: “points that are close in X should map to points close in Y ”



Gromov-Wasserstein: Discrete Case